



Strathmore
UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES (SCES)
BACHELORS OF SCIENCE IN COMPUTER NETWORKS AND CYBER SECURITY
CONTINUOUS ASSESSMENT TEST (CAT 2)
CNS 2205: NETWORK PROGRAMMING

DATE: 17 November 20205

Time: 1 Hour

Instructions

1. This CAT consists of **30** multiple choice questions.
2. Attempt ALL questions.

1. Which of the following UDP ports does SNMP use by default?
 - A. 22
 - B. 80
 - C. 161**
 - D. 443
2. Which component of SNMP stores information about managed devices?
 - A. OID
 - B. MIB**
 - C. Agent
 - D. Trap
3. What is an SNMP agent?
 - A. A central management station
 - B. A device component that collects and reports information**
 - C. A type of network administrator
 - D. A secure communication protocol
4. Which SNMP operation allows modifying a variable on the agent?
 - A. GETNEXT
 - B. SET**

- C. INFORM
- D. TRAP

5. The SNMP manager's main function is to:
 - A. Store network topology data
 - B. Poll agents and process responses
 - C. Generate routing tables
 - D. Filter packets
6. What is Nagios primarily used for?
 - A. Web hosting
 - B. Network and system monitoring
 - C. Data encryption
 - D. Software development
7. Which command is used to check the syntax of Nagios configuration files?
 - A. nagios --check
 - B. nagios -v /etc/nagios/nagios.cfg
 - C. systemctl verify nagios
 - D. nagios --syntax
8. Which file defines Nagios hosts, services, and contacts?
 - A. nagios.cfg
 - B. objects.cfg
 - C. commands.cfg
 - D. nagios.conf
9. What is NRPE used for in Nagios?
 - A. Network routing
 - B. Remote plugin execution
 - C. Plugin compilation
 - D. Log storage
10. What is the main difference between "active" and "passive" checks in Nagios?
 - A. Passive checks require plugins; active checks don't
 - B. Active checks are triggered by Nagios; passive checks are submitted by external processes
 - C. Passive checks use NRPE only
 - D. There is no difference
11. What is iptables used for in Linux?
 - A. Monitoring system performance
 - B. Filtering and controlling network traffic

- C. Configuring file permissions
- D. Encrypting files

12. Which kernel framework does iptables use for packet filtering?

- A. ebttables
- B. nftables
- C. Netfilter
- D. Firewallld

13. Which chain in iptables is used for packets initiated locally on the host?

- A. INPUT
- B. OUTPUT
- C. FORWARD
- D. PREROUTING

14. Which iptables command lists all the current firewall rules?

- A. iptables -L
- B. iptables -A
- C. iptables -F
- D. iptables -D

15. What is the default action if a packet does not match any rule?

- A. DROP
- B. ACCEPT
- C. REJECT
- D. It depends on the policy of the chain

16. Which target is used to silently discard a packet?

- A. DROP
- B. REJECT
- C. LOG
- D. RETURN

17. The -A option in iptables is used to:

- A. Append a rule to a chain
- B. Add a new table
- C. Apply NAT
- D. Allow all traffic

18. Which of the following tables is used for packet alteration (e.g., TTL or QoS marking)?

- A. nat
- B. filter
- C. mangle
- D. raw

19. Which table in iptables is used for setting up connection tracking exemptions?
- A. filter
 - B. raw
 - C. mangle
 - D. security
20. To flush (delete) all rules in iptables, the command is:
- A. iptables -D
 - B. iptables -Z
 - C. iptables -F
 - D. iptables -X
21. Which of the following commands allows SSH traffic (port 22) to the local system?
- A. iptables -A INPUT -p udp --dport 22 -j ACCEPT
 - B. iptables -A OUTPUT -p tcp --dport 22 -j DROP
 - C. iptables -A FORWARD -p tcp --dport 22 -j REJECT
 - D. iptables -A INPUT -p tcp --dport 22 -j ACCEPT
22. What is the main purpose of the POSTROUTING chain in the nat table?
- A. Handle incoming packets
 - B. Modify packets before routing
 - C. Modify packets after routing (e.g., SNAT/MASQUERADE)
 - D. Drop packets after routing
23. Which iptables chain is used to alter destination IP addresses of incoming packets?
- A. PREROUTING
 - B. INPUT
 - C. OUTPUT
 - D. FORWARD
24. Which chain is used for Source NAT (SNAT) in iptables?
- A. PREROUTING
 - B. POSTROUTING
 - C. INPUT
 - D. FORWARD
25. Which of the following commands performs SNAT for all outgoing packets from 192.168.1.0/24 to 203.0.113.5?
- A. iptables -t nat -A PREROUTING -s 192.168.1.0/24 -j SNAT --to-source 203.0.113.5
 - B. iptables -t nat -A POSTROUTING -s 192.168.1.0/24 -j SNAT --to-source 203.0.113.5
 - C. iptables -t nat -A FORWARD -s 192.168.1.0/24 -j MASQUERADE
 - D. iptables -t nat -A OUTPUT -s 192.168.1.0/24 -j SNAT
26. Which of the following is used for port forwarding to an internal web server (10.0.0.2:80)?

- A. `iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to 10.0.0.2:80`
- B. `iptables -t nat -A POSTROUTING -p tcp --dport 80 -j SNAT --to 10.0.0.2:80`
- C. `iptables -t nat -A OUTPUT -p tcp --dport 80 -j MASQUERADE`
- D. `iptables -t mangle -A PREROUTING -j DNAT --to 10.0.0.2:80`

27. Which of the following commands enables masquerading on the interface eth0?

- A. `iptables -t nat -A PREROUTING -o eth0 -j MASQUERADE`
- B. `iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE`
- C. `iptables -A INPUT -i eth0 -j MASQUERADE`
- D. `iptables -t nat -A FORWARD -o eth0 -j SNAT`

28. To remove all NAT rules from the NAT table, which command is correct?

- A. `iptables -X nat`
- B. `iptables -F`
- C. `iptables -t nat -F`
- D. `iptables -N nat`

29. Which of the following runs in kernel space?

- A. Shell
- B. Text editor
- C. Device driver
- D. Web browser

30. What is the purpose of the “system call interface” in the Linux kernel?

- A. To allow communication between two kernels
- B. To allow user applications to request services from the kernel
- C. To manage device drivers
- D. To compile kernel source code